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Learning Objectives of This Lecture

1. Be familiar with the judgment and period calculation of discrete periodic signals;
2. Be familiar with the basic operations of sequences;
3. Be able to establish the difference equation of discrete-time systems;
4. Be familiar with the classical time-domain solution method of difference equations and correctly use **boundary conditions**.

Continuous-Time Signals:

$f(t)$ is a continuously changing function of t , and a definite function value can be given for any time value t except for several discontinuous points. The waveform of the signal is a smooth curve, and it is generally also called an analog signal.

Continuous-Time Systems:

The input and output of the system are both continuous-time signals, e.g., physics, modern circuit theory, and analog communication systems.

True or False: The signal obtained by sampling a continuous-time signal is a digital signal.

☐ A True

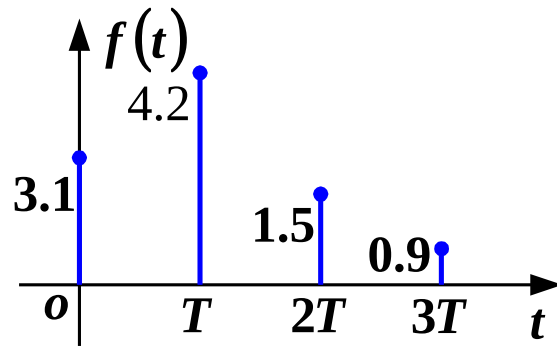
☒ B False

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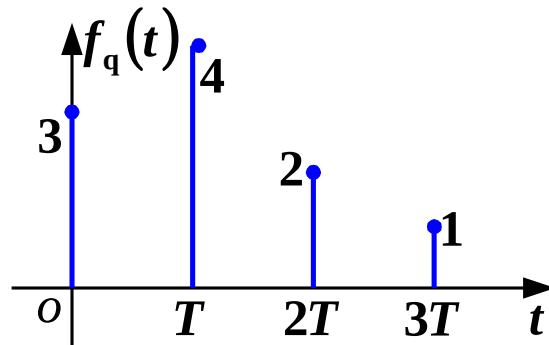
6.1 Introduction



Discrete-Time Signal: The time variable is discrete, and the function has definite values only at certain specified moments, being undefined at other times. It can be obtained by sampling an analog signals or generated by an actual system.



Sampling Process — Obtains discrete-time signals.



Amplitude Quantization — The amplitude is of limited levels.

Digital Signal: A signal where the amplitude of a discrete-time signal is quantized at each discrete point.

6.1 Introduction

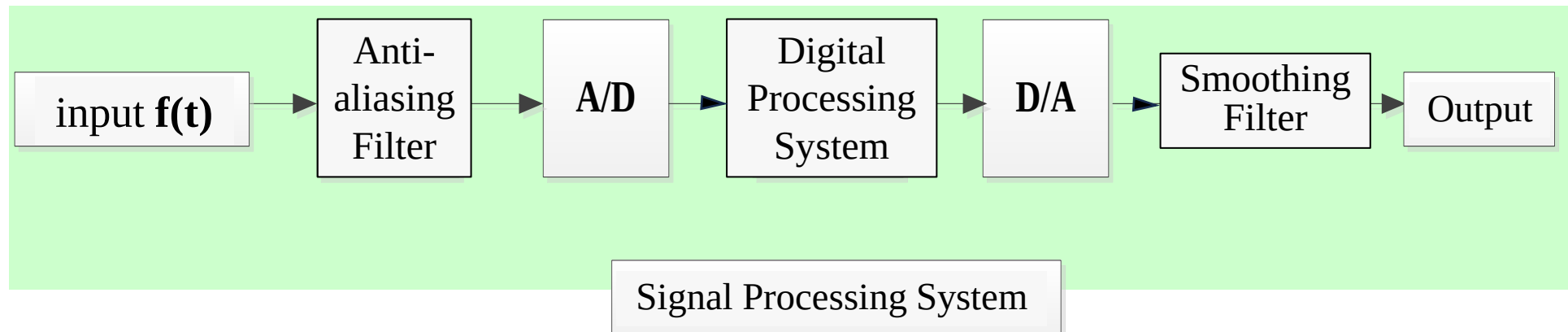


Discrete-Time System: Both the input and output of the system are discrete-time signals. Examples include computer, numerical analysis, statistics, and economics.

Advantages of Discrete-Time Systems:

1. Facilitate large-scale integration, featuring small size, light weight, and low power consumption;
2. High reliability and high precision (depending on the number of bits of binary numbers);
3. Store information using memory, with flexible functions;
4. Easy to eliminate noise and process low-frequency signals;
5. Process multi-dimensional signals;
6. Programmable technology improves flexibility and versatility.

Hybrid System: The combined application of continuous-time systems and discrete-time systems. Examples include automatic control systems and digital communication systems.



Continuous-Time System

- Differential Equation
 - Convolution Integral
 - Laplace Transform
 - Fourier Transform
-
- System Function
 - Convolution Theorem

vs.

Discrete-Time System

- Difference Equation
- Convolution Sum
- z-Transform
- Discrete-Time Fourier Transform, Discrete Fourier Transform
- System Function
- Convolution Theorem

6.2 Discrete-Time Signals – Sequences



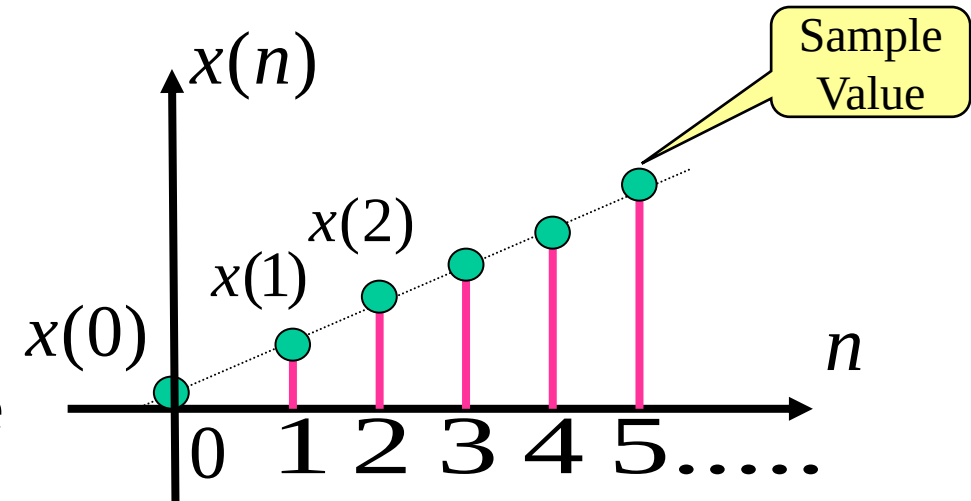
Discrete-Time Signal (Sequence): It is discrete in time, with function values defined only at certain discontinuous specified moments and undefined at other times.

The interval between discrete moments is uniform, denoted as T .

Discrete-time signals are denoted as $\{x(nT)\}$ or $\{x(n)\}$, $n \in \mathbb{Z}, -\infty < n < \infty$

For simplicity, the sequence is denoted as $x(n)$.

- Listing values, e.g., $x(n)=\{1,2,3\}$
- Closed-form solution, e.g., $x(n)=a^n u(n)$
- Graphical representation, as shown in the figure

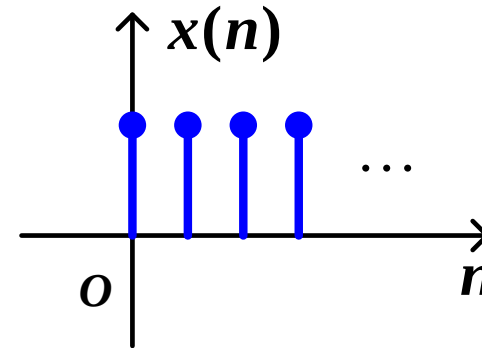


6.2 Discrete-Time Signals - Sequences

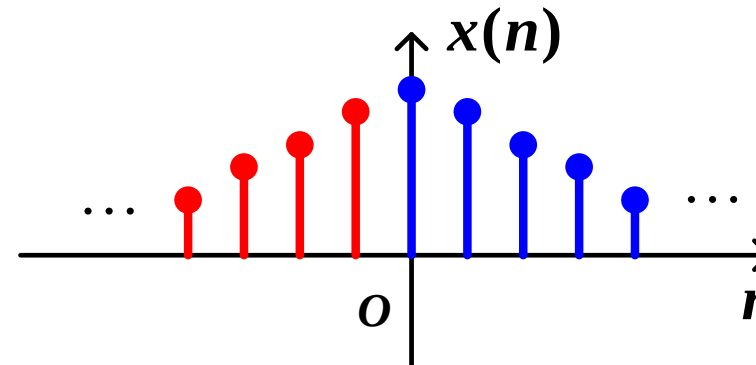


Three Forms of Sequences:

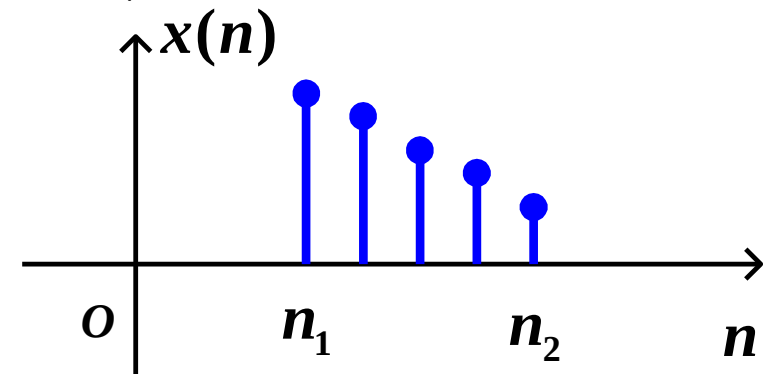
Single-sided (Causal) Sequence: $n \geq 0$;



Double-sided Sequence: $-\infty \leq n \leq \infty$;



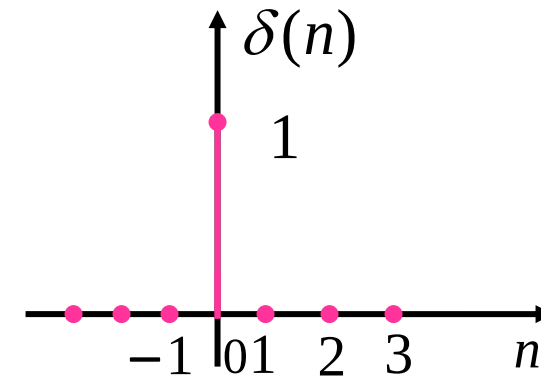
Finite-Length Sequence: $n_1 \leq n \leq n_2$.



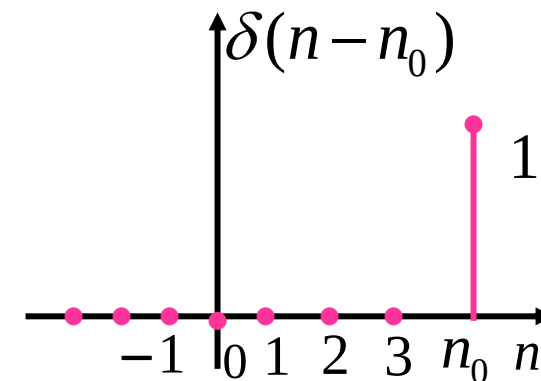
6.2.1 Commonly Used Typical Sequences

1. Unit Sample/ Unit Impulse

$$\delta(n) = \begin{cases} 1 & (n = 0) \\ 0 & (n \neq 0) \end{cases}$$



$$\delta(n - n_0) = \begin{cases} 1 & (n = n_0) \\ 0 & (n \neq n_0) \end{cases}$$



Note: $\delta(t)$ is represented by area (intensity)(as $t \rightarrow 0$, the amplitude is ∞);
 $\delta(n)$ takes a finite amplitude of 1 at $n=0$ (not area).

6.2 Discrete-Time Signals - Sequences

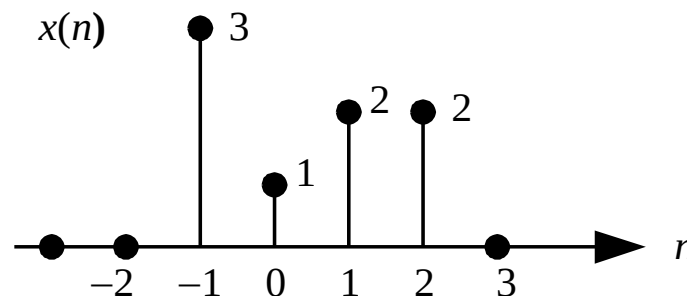


Representing Any Discrete-Time Signal Using Unit Sample Sequences

$$x(n) = \sum_{m=-\infty}^{\infty} x(m)\delta(n-m)$$

$$\text{where } x(m)\delta(n-m) = \begin{cases} x(n) & m = n \\ 0 & m \neq n \end{cases}$$

Any sequence is the sum of weighted and delayed unit sample sequences.



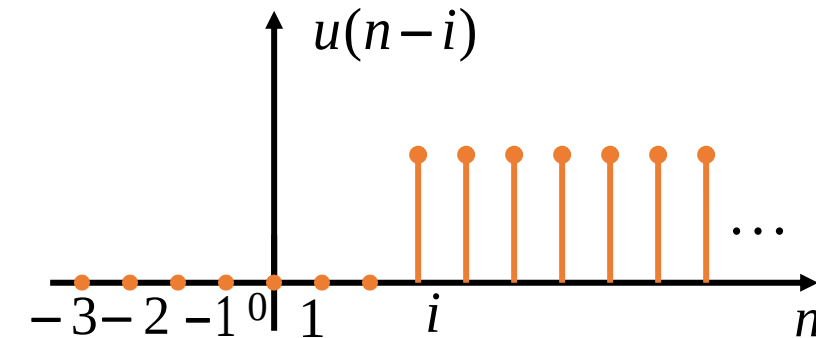
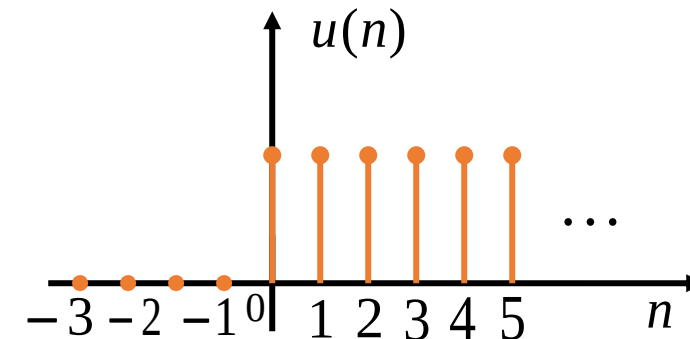
$$x(n) = 3\delta(n+1) + \delta(n) + 2\delta(n-1) + 2\delta(n-2)$$

2. Unit Step Sequence

$$u(n) = \begin{cases} 1 & (n \geq 0) \\ 0 & (n < 0) \end{cases}$$

$$u(n-i) = \begin{cases} 1 & (n \geq i) \\ 0 & (n < i) \end{cases}$$

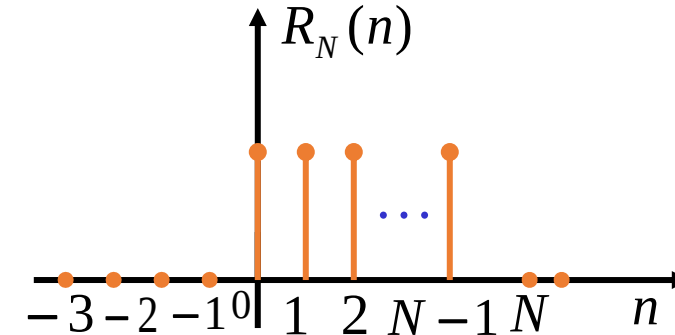
$$\left\{ \begin{array}{l} u(n) = \sum_{K=0}^{\infty} \delta(n-K) \\ \delta(n) = u(n) - u(n-1) \end{array} \right.$$



$\delta(n)$ and $u(n)$ have a difference – sum relationship, not a calculus relationship anymore.

3. Rectangular Sequence

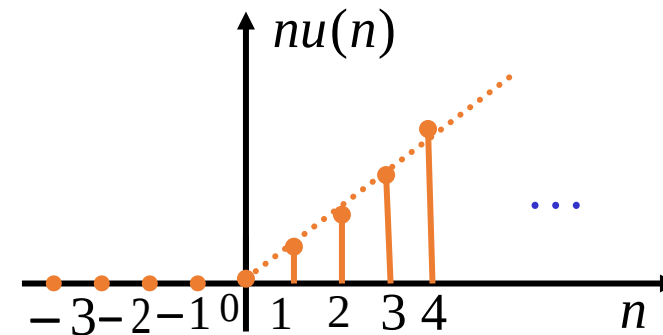
$$R_N(n) = \begin{cases} 1 & (0 \leq n \leq N-1) \\ 0 & (n < 0, n \geq N) \end{cases}$$



$$R_N(n) = u(n) - u(n-N) = \sum_{k=0}^{N-1} \delta(n-k)$$

4. Ramp Sequence

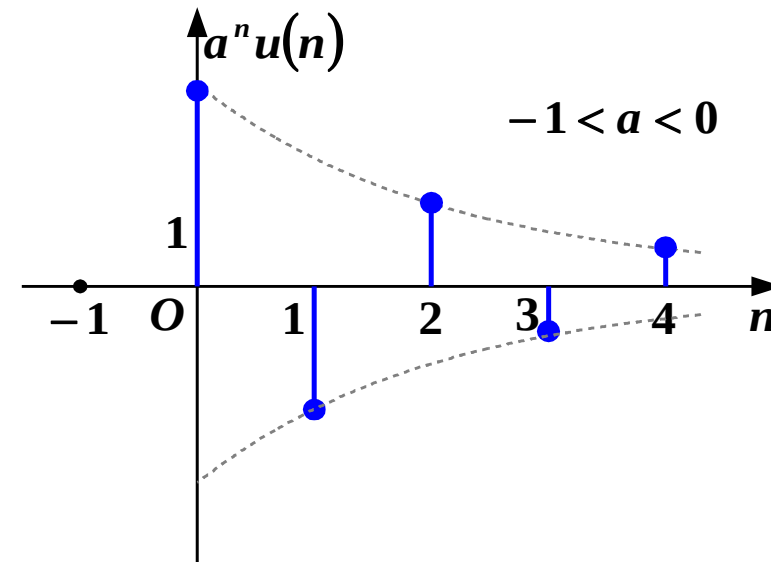
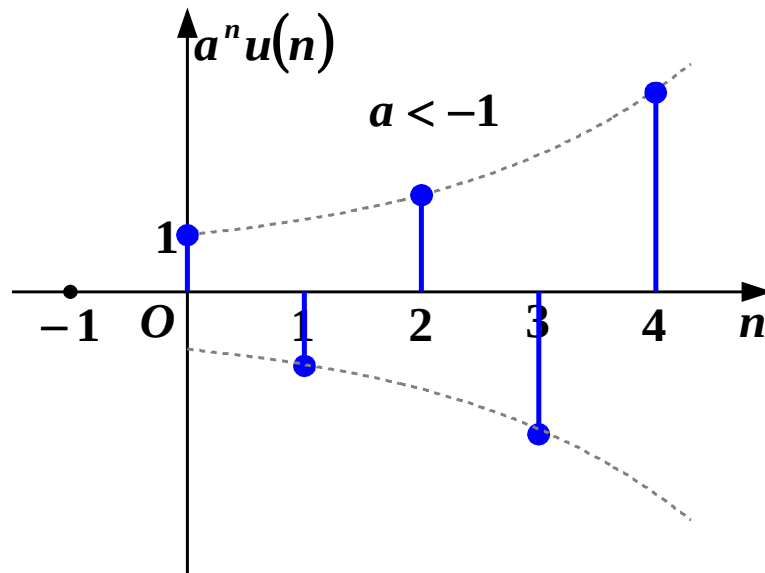
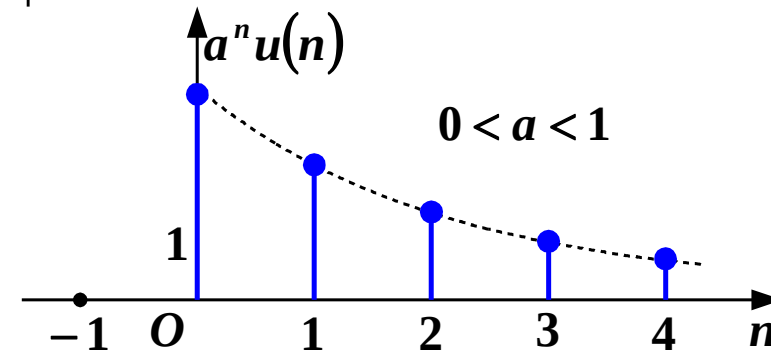
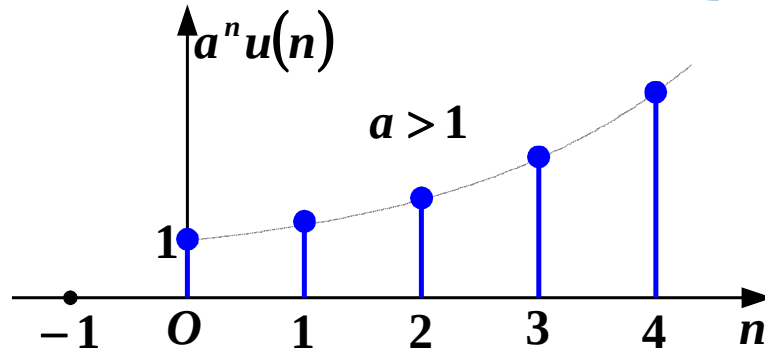
$$x(n) = nu(n)$$



5. Real-Valued Exponential Sequence

$$x(n] = a^n u(n)$$

{ when $|a| > 1$ the sequence diverges;
when $|a| < 1$ the sequence converges.

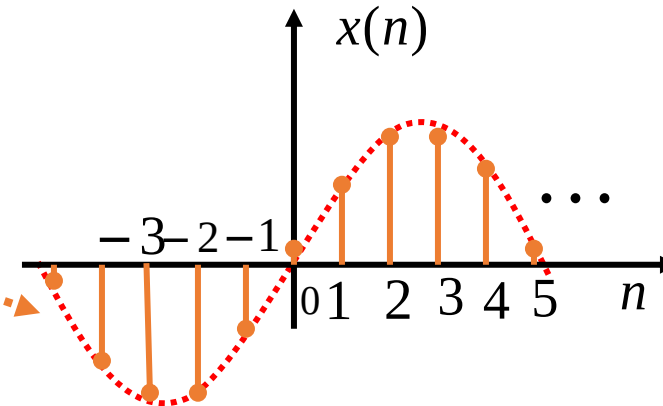


6. Sinusoidal Sequence

$$f(t) = \sin \Omega_0 t$$

$$x(n) = f(nT)$$

$$= \sin(\Omega_0 nT) = \sin(n\omega_0)$$



where ω_0 is the frequency of the sinusoidal sequence, reflecting the rate at which the sequence values repeat periodically in turn. For example, $\omega_0 = 2\pi / 10$.

$$\omega_0 = \Omega_0 T = \frac{\Omega_0}{f_s}$$

ω_0 is the normalized value of Ω_0 with respect to f_s

Is the sequence $\sin \frac{4\pi}{11}n$ a periodic signal? If yes, what is its period?

- ☐ A No
- ☐ B Yes, $11/4$
- ☐ C Yes, $11/2$
- ☒ D Yes, 11

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6.2 Discrete-Time Signals – Sequences



Determination of Periodicity of Sine Sequences:

① $\frac{2\pi}{\omega_0} = N$, N is a positive integer

$$\sin[\omega_0(n + N)] = \sin\left[\omega_0\left(n + \frac{2\pi}{\omega_0}\right)\right] = \sin(\omega_0 n + 2\pi) = \sin(\omega_0 n)$$

The sine sequence is periodic, **with period N**

② $\frac{2\pi}{\omega_0} = \frac{N}{m}$, $\frac{N}{m}$ is a rational number

$$\sin[\omega_0(n + N)] = \sin\left[\omega_0\left(n + m\frac{2\pi}{\omega_0}\right)\right] = \sin(\omega_0 n + m \cdot 2\pi) = \sin(\omega_0 n)$$

$\sin(\omega_0 n)$ is still periodic. **Period:** $m\frac{2\pi}{\omega_0} = N$

③ $\frac{2\pi}{\omega_0}$ is an irrational number

There is no value of N that satisfies $x(n + N) = x(n)$, so the sequence is **aperiodic**.

6.2 Discrete-Time Signals – Sequences

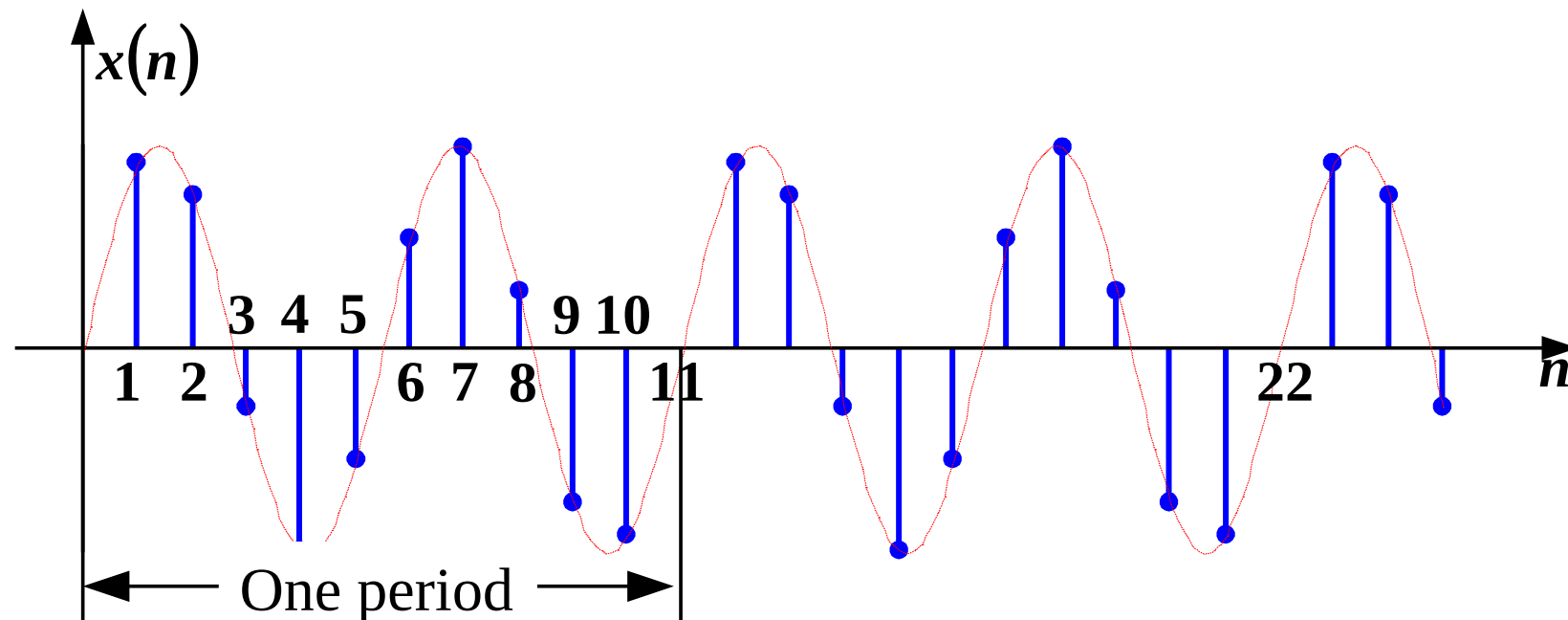


Example 6-1: Given $\sin \frac{4\pi}{11} n$, find its period.

Solution:

$$\omega_0 = \frac{4\pi}{11} \quad , \quad \text{then} \quad \frac{2\pi}{\omega_0} = 2\pi \frac{11}{4\pi} = \frac{11}{2} = \frac{N}{m}$$

so $N = 11$, that is, the period is **11**. (There are 5.5 ω_0 in 2π)



7. Complex Exponential Sequence

$$x(n) = e^{j\omega_0 n} = \cos(\omega_0 n) + j \sin(\omega_0 n)$$

A complex sequence can be expressed in polar coordinates:

$$x(n) = |x(n)| e^{j \arg[x(n)]}$$

$$|x(n)| = 1 \quad \arg[x(n)] = \omega_0 n$$

General complex exponential sequence

$$x(n) = Ae^{(\sigma + j\omega_0)n} = Ae^{\sigma n} e^{j\omega_0 n}$$

$$|x(n)| = Ae^{\sigma n} \quad \arg[x(n)] = \omega_0 n$$

$$x(n) = Ae^{\sigma n} e^{j\omega_0 n} = Ae^{\sigma n} \cos(\omega_0 n) + jAe^{\sigma n} \sin(\omega_0 n)$$

6.2 Discrete-Time Signals – Sequences



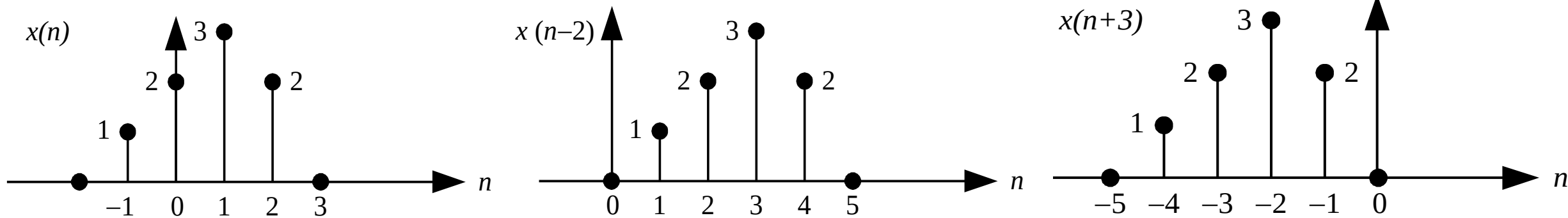
6.2.2 Operations on Discrete-Time Signals

1. Shifting of Sequences

$x(n-m)$ means shifting $x(n)$ to the right by m units.

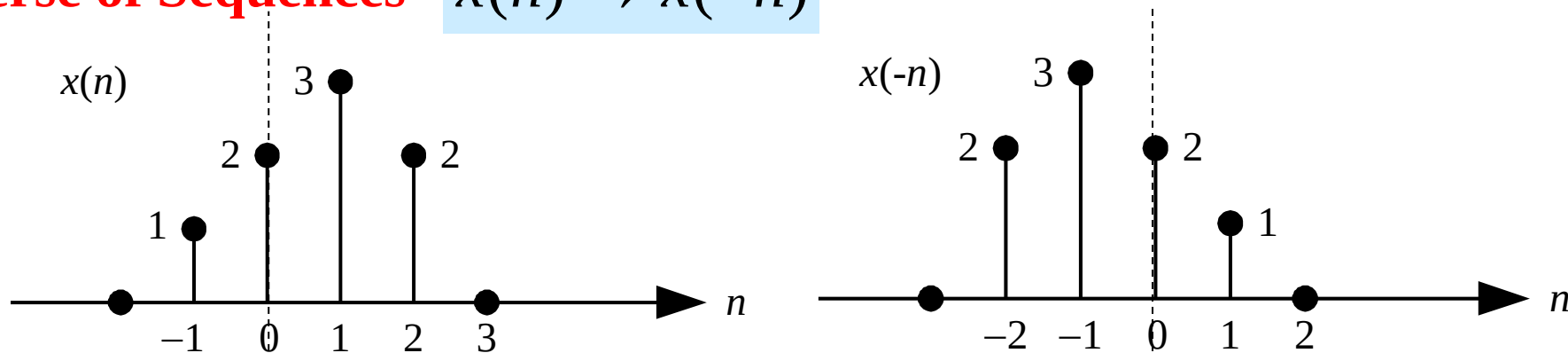
$x(n+m)$ means shifting $x(n)$ to the left by m units.

$m > 0$



2. Inverse of Sequences

$x(n) \rightarrow x(-n)$



3. Scaling of Sequences

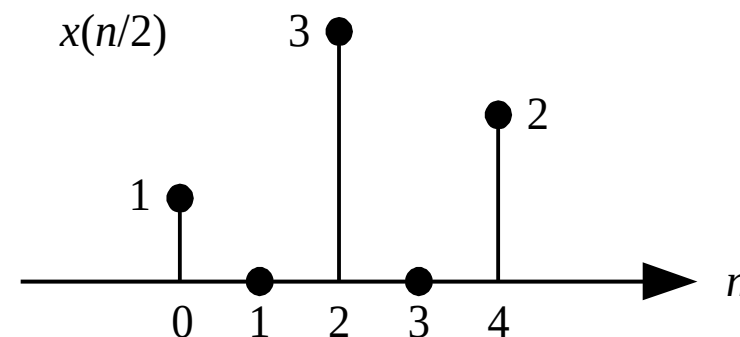
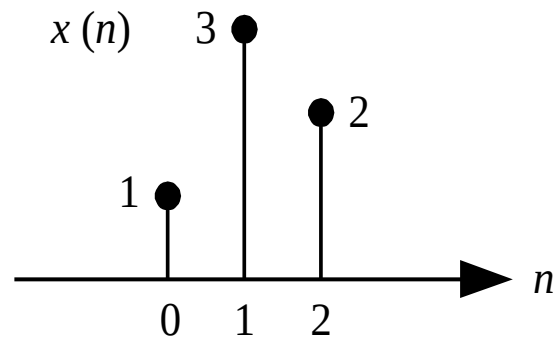
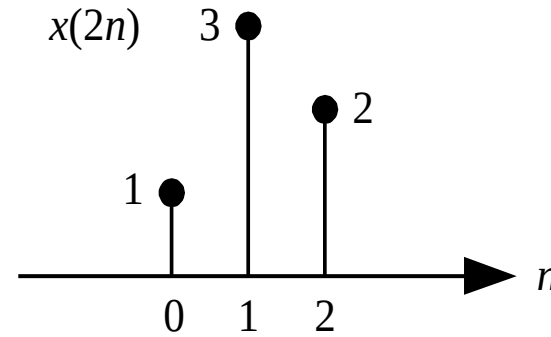
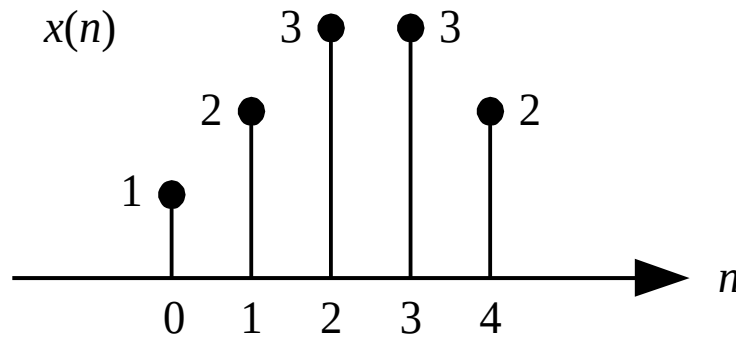
$a > 0$ is a positive integer.

$$x(n) \rightarrow x(an)$$

$$x(n) \rightarrow x\left(\frac{n}{a}\right)$$

Compression: Extract one point every $(a-1)$ points around the origin.

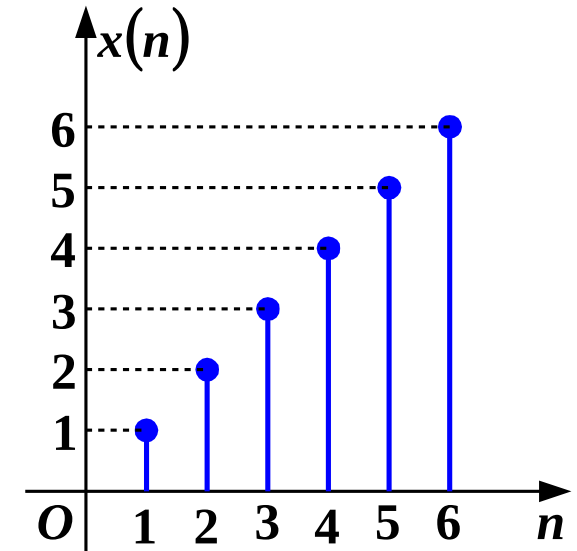
Expansion: Insert $(a-1)$ zero-valued points between adjacent points.



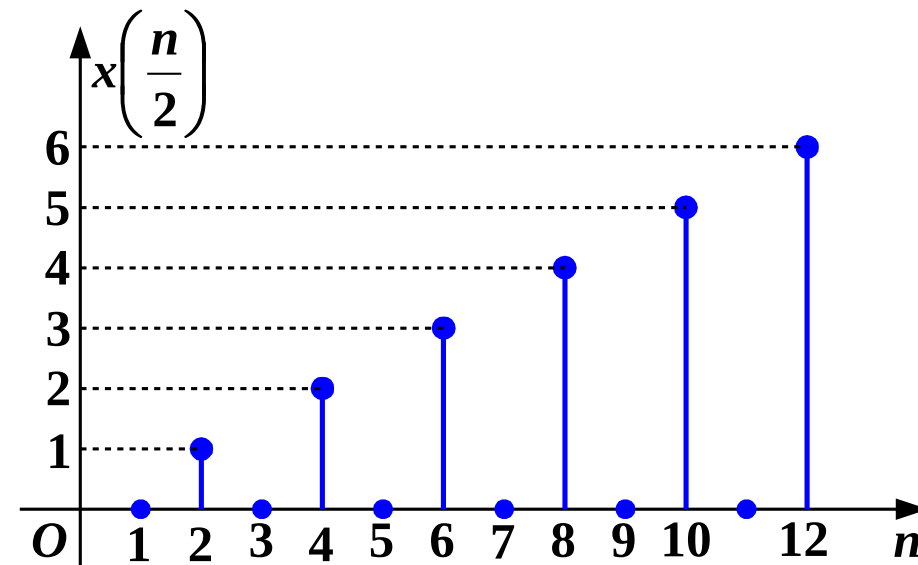
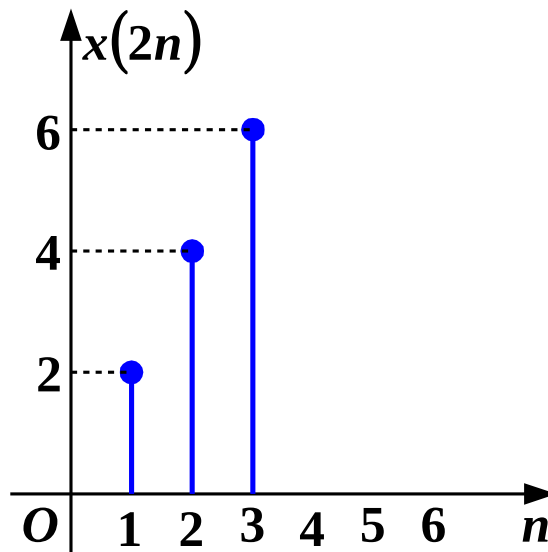
6.2 Discrete-Time Signals – Sequences



Example 6-2: Given the waveform of $x(n]$, please draw the waveforms of $x(2n]$ and $x(n/2]$.



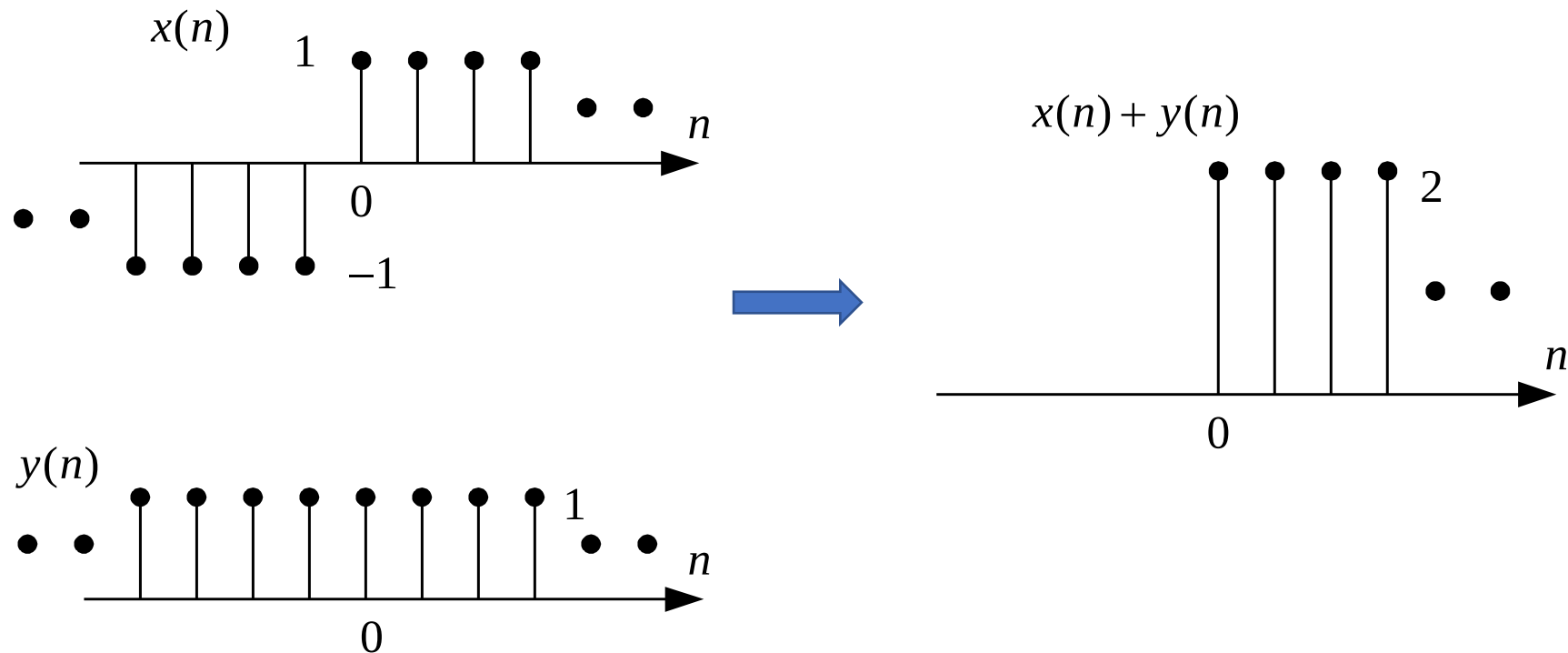
Solution:



4. Addition of Sequences

$$z(n) = x(n) + y(n)$$

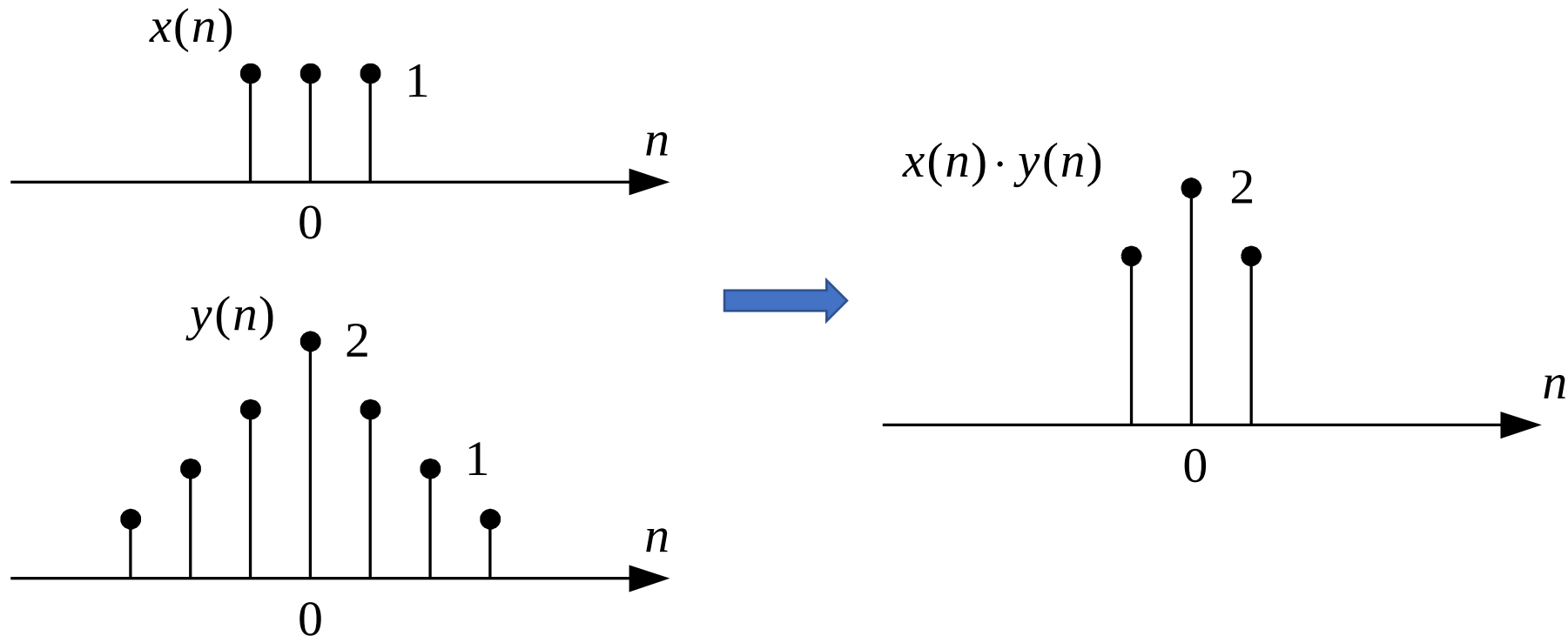
Add term by term.



5. Multiplication of Sequences

$$z(n) = x(n) \cdot y(n)$$

Multiply term by term.



6. Difference of Sequences

Backward Difference: Subtract the past value from the current value.

First-order backward difference $\nabla x(n) = x(n) - x(n-1)$

Second-order backward difference $\nabla^2 x(n) = \nabla\{\nabla x(n)\}$
$$= [x(n) - x(n-1)] - [x(n-1) - x(n-2)]$$
$$= x(n) - 2x(n-1) + x(n-2)$$

Forward Difference: Subtract the current value from the future value.

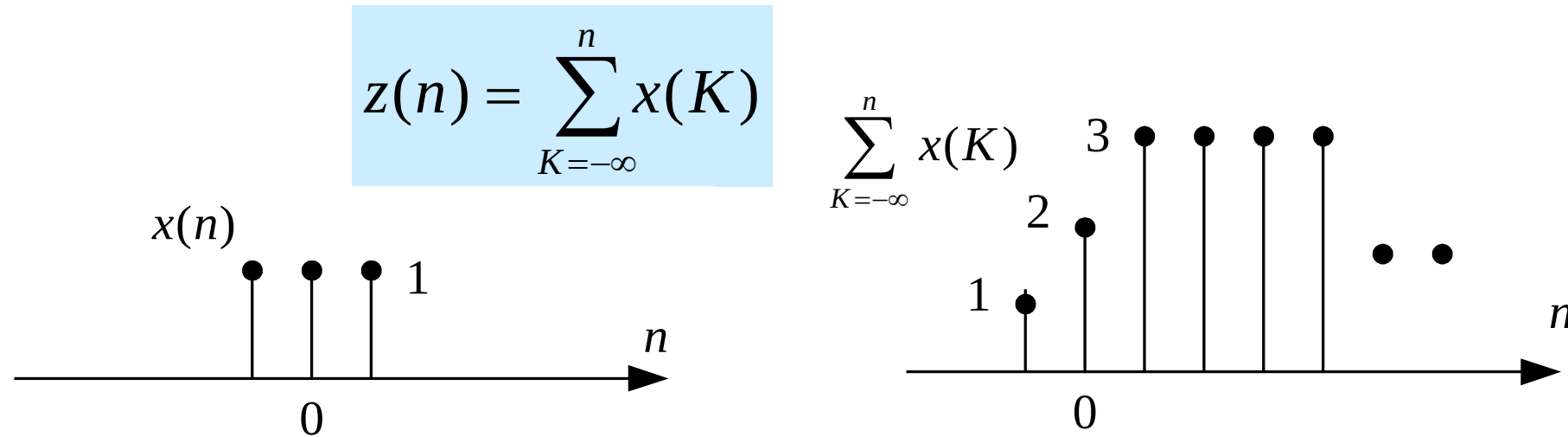
First-order forward difference $\Delta x(n) = x(n+1) - x(n)$

Second-order forward difference $\Delta^2 x(n) = \Delta\{\Delta x(n)\} = x(n+2) - 2x(n+1) + x(n)$

The unit sample sequence can be expressed by the difference of the unit step sequence:

$$\delta(n) = u(n) - u(n-1)$$

7. Summation of Sequences



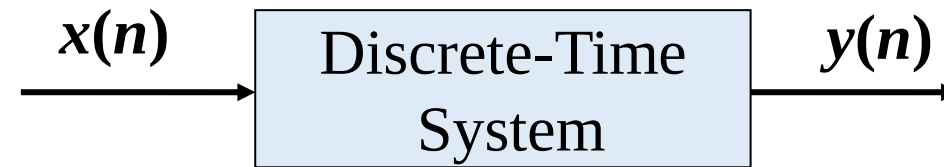
The unit step sequence can be expressed by the summation of the unit sample sequence:

$$u(n) = \sum_{K=-\infty}^n \delta(K)$$

8. Energy of a Sequence

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

6.3.1 Discrete Linear Time-Invariant System

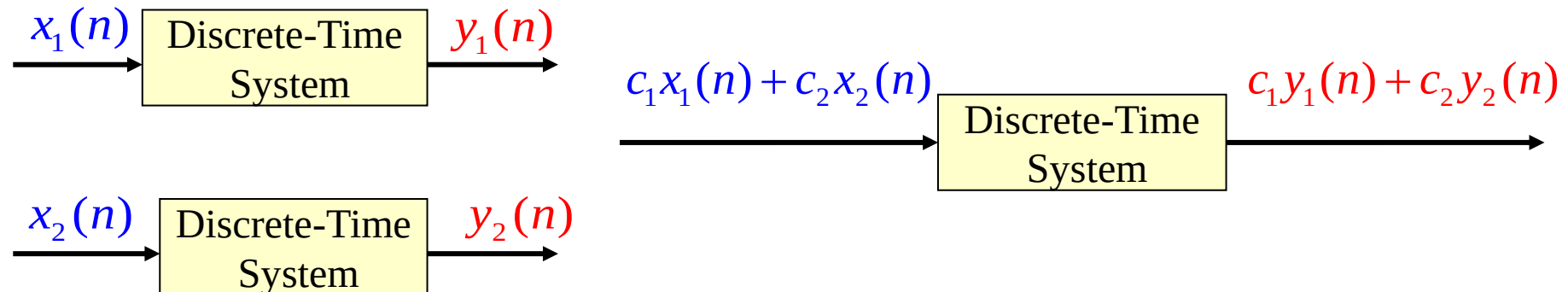


1. Linearity

Homogeneity and Superposition: Suppose two pairs of excitation and response

$$x_1(n) \rightarrow y_1(n), x_2(n) \rightarrow y_2(n)$$

$$c_1 x_1(n) + c_2 x_2(n) \rightarrow c_1 y_1(n) + c_2 y_2(n)$$

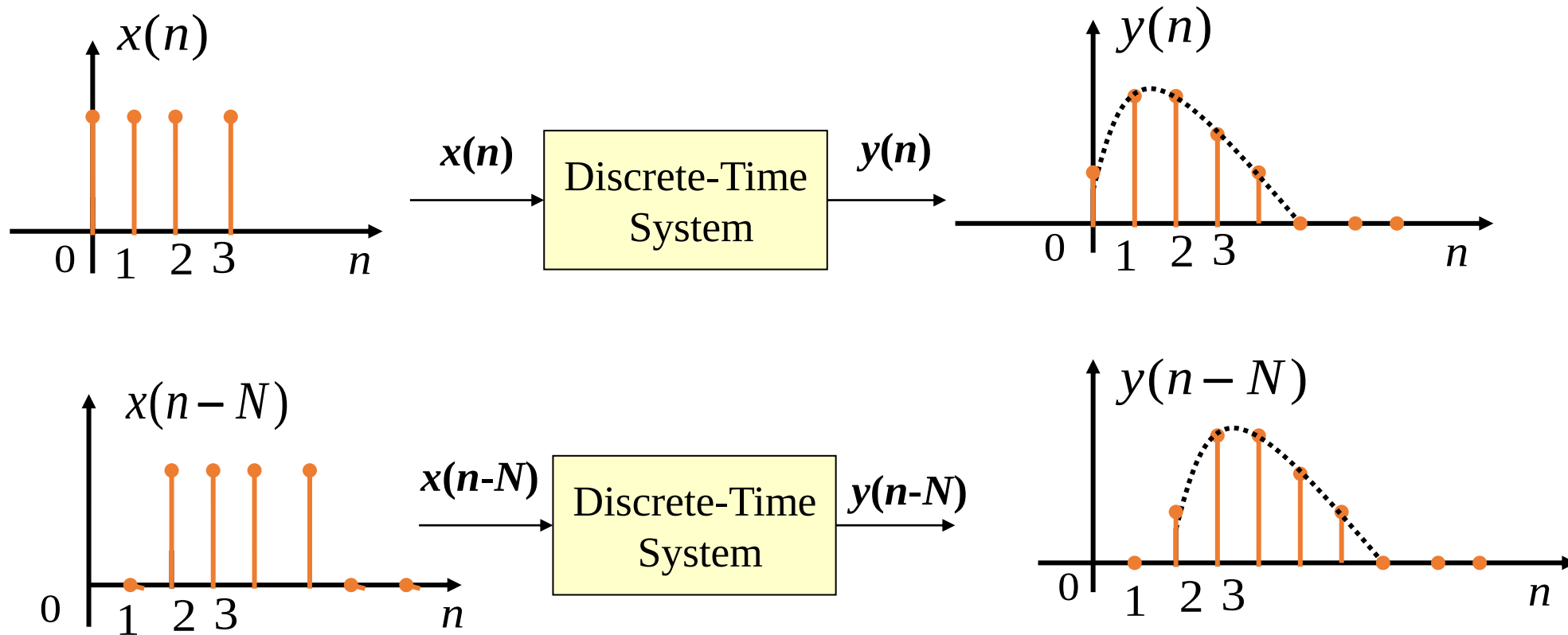


6.3 Mathematical Model of Discrete-Time Systems



2. Time-Invariance

Suppose the excitation and response satisfy $x(n) \rightarrow y(n)$, then $x(n - N) \rightarrow y(n - N)$.



6.3.2 Mathematical Model of Discrete-Time Systems

Mathematical Model of Continuous Systems—Differential Equation

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + E_m e(t)$$

Basic Operations: Differentiation, Multiplication by Coefficients, Addition

Mathematical Model of Discrete Systems—Difference Equation

$$y(n) = -\sum_{k=1}^N a_k y(n-k) + \sum_{r=0}^M b_r x(n-r)$$

The input is the discrete sequence $x(n)$ and its time-shifted functions
 $x(n), x(n-1), x(n-2), \dots$

The output is the discrete sequence $y(n)$ and its time-shifted functions

$$y(n), y(n-1), y(n-2), \dots$$

Basic Operations: **Delay**, Multiplication by Coefficients, Addition

6.3.3 Basic Operational Units of Discrete Systems

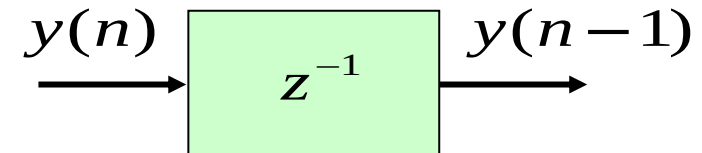
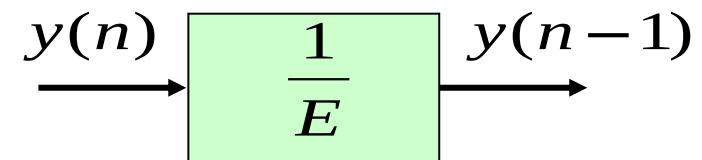
Delay Element (Unit Delay)

For LTI systems, concepts such as operator symbols and transfer operators can be used to represent or solve the mathematical model of the system. For discrete-time systems, the operator symbol E is used to denote the operation of advancing a sequence by one unit of time. E is also called a time-shift operator, and using the shift operator, we can write:

$$y(n+1) = Ey(n) \quad y(n-1) = \frac{1}{E} y(n)$$

The above analysis shows that the operator $\frac{1}{E}$ represents the function of delaying by one unit of time, that is, $y(n)$ undergoes the $\frac{1}{E}$ operation to give $y(n-1)$. It is specified that $\frac{1}{E}$ is used as the symbol of the **delay element**.

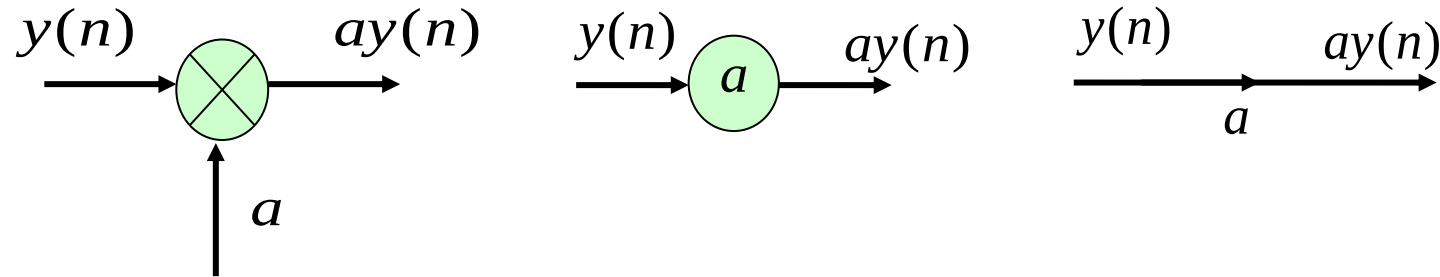
A unit delay is actually a **shift register**, which pushes out the previous discrete value and replenishes it.



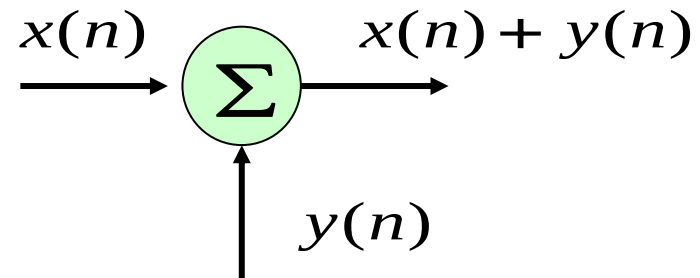
6.3 Mathematical Model of Discrete-Time Systems



Multiplier



Adder



6.3.5 Characteristics of Difference Equations

1. The n -th value of the output sequence is determined not only by the input sample at the same instant but also by the previous output values, and **each output value must be retained sequentially**.
2. Order of Difference Equation: The order is the difference between the highest and lowest indices of variables in the difference equation. **If the n -th output of a system is determined by several just-past output values and input values, the difference equation describing it is of that order.**

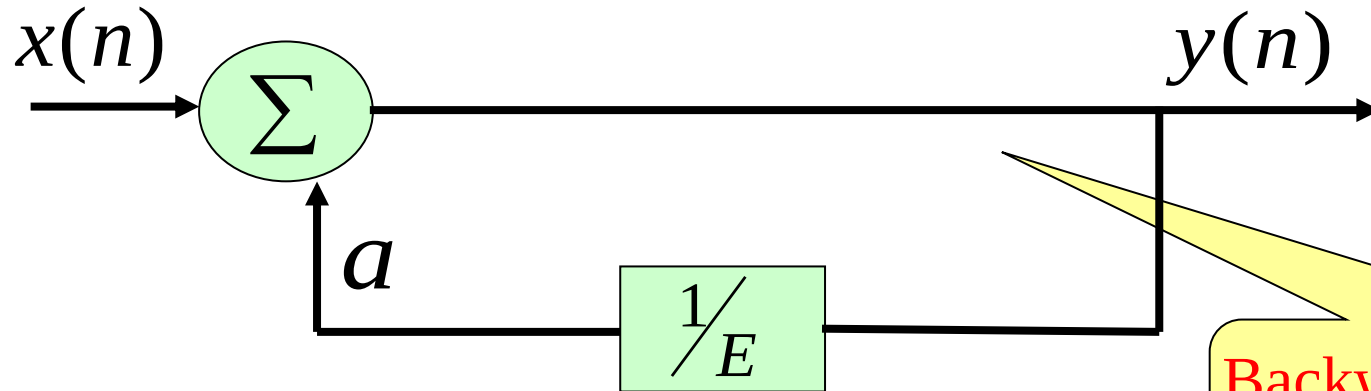
General Form:
$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

3. Differential equations can be approximated by difference equations. The solution of a differential equation is an exact solution, while the solution of a difference equation is an approximate solution, and they have many similarities.
4. Difference equations describe discrete-time systems. The operational relationship between the input sequence and the output sequence corresponds to the **system simulation block diagram**, and one should be able to write and draw it.

6.3 Mathematical Model of Discrete-Time Systems

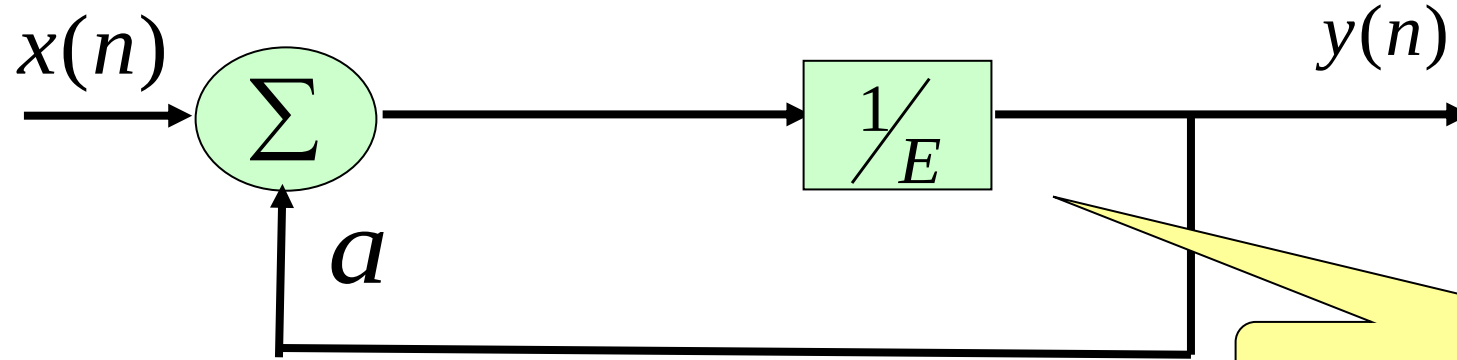


There is no essential difference between the two systems !



$$y(n) = x(n) + ay(n-1)$$

Backward Difference Equation
(More Commonly Used)



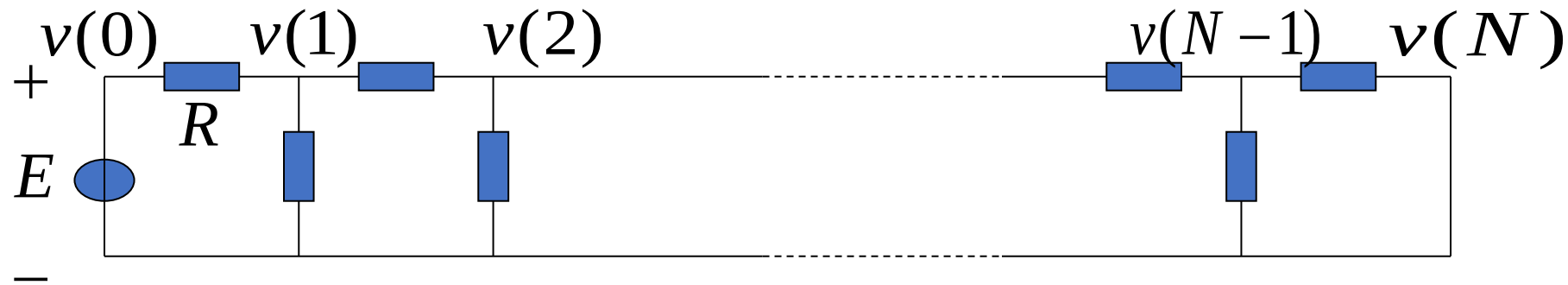
$$y(n+1) = ay(n) + x(n)$$

Forward Difference Equation

6.3 Mathematical Model of Discrete-Time Systems



Example 6-3: A resistive ladder network, where each branch resistance is R , The voltage of each node to ground is $v(n)$, $n=0, 1, 2, \dots, N$. Given the boundary node voltages $v(0)=E$, $v(N)=0$. derive the difference equation for $v(n)$.



Solution:

$$\frac{v(n-1)}{R} = \frac{v(n) - v(n-1)}{R} + \frac{v(n-2) - v(n-1)}{R}$$

$$v(n) - 3v(n-1) + v(n-2) = 0$$

6.4 Classical Time-Domain Solution of Linear Difference Equations

Linear time-invariant discrete systems are described by linear difference equations with constant coefficients:

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

The order of the equation is equal to the difference between the highest and lowest indices of the unknown sequence variables.

Solving Methods:

- **Iteration Method**
- **Classical Time-Domain Method:** Find Homogeneous Solution and Particular Solution
- **Convolution Method:** Use the homogeneous solution to get the zero-input response, and then use the Convolution Sum to find the zero-state response.
- **Transform Domain Method** (z-Transform Method, Chapter 7)
- **State Variable Analysis Method** (Chapter 8)

6.4 Classical Time-Domain Solution of Linear Difference Equations

The complete solution of a difference equation, i.e., the complete response of the system, consists of the homogeneous solution $y_h(n)$ and the particular solution $y_p(n)$:

$$y(n) = y_h(n) + y_p(n)$$

The form of the homogeneous solution $y_h(n)$ is determined by the characteristic roots of the homogeneous equation.

The form of the particular solution $y_p(n)$ is determined by the simplified form of the right-hand side of the equation after substituting the excitation signal.

1. Find the Homogeneous Solution (Inherent Response of the System)

Homogeneous equation of the difference equation:: $\sum_{k=0}^N a_k y(n-k) = 0$

$$a_0 \alpha^N + a_1 \alpha^{N-1} + \cdots + a_k \alpha^{N-k} + \cdots + a_N = 0$$

Characteristic Equation

The roots of the equation $\alpha_1, \alpha_2, \alpha_3 \dots$

Characteristic Roots

α_i is called the natural frequency (or free frequency) of the system.

6.4 Classical Time-Domain Solution of Linear Difference Equations

Homogeneous Solutions of Difference and Differential Equations

Characteristic Roots	Homogeneous Solution of Difference Equation	Homogeneous Solution of Differential Equation
Distinct Real Roots $\alpha_1, \alpha_2, \dots, \alpha_N$	$\sum_{k=1}^N C_k \alpha_k^n$	$\sum_{k=1}^N A_k e^{\alpha_k t}$
k -fold Root α_1	$\sum_{i=1}^k C_i n^{k-i} \alpha_1^n$	$\sum_{i=1}^k A_i t^{k-i} e^{\alpha_1 t}$
Conjugate Complex Roots $\alpha \pm j\beta$	$\left(\sqrt{\alpha^2 + \beta^2} \right)^n \left[P \cos(n\omega_0) + Q \sin(n\omega_0) \right]$ $\rho = \sqrt{\alpha^2 + \beta^2} \quad P = C_1 + C_2 \quad Q = j(C_1 - C_2)$	$e^{\alpha t} (A \cos \beta t + B \sin \beta t)$

The undetermined coefficients of the homogeneous solution are determined by the **boundary conditions** (after finding the particular solution).

6.4 Solution of Linear Constant-Coefficient Difference Equations

2. Find the Particular Solution: Determined by the Form of the Excitation Signal

Free Term (Right-Hand Side of the Equation)	Form of Particular Solution
α^n	$D\alpha^n$ (if α is not a characteristic root) $Dn^r\alpha^n$ (if α is an r -fold characteristic root)
n^k	$D_0n^k + \dots + D_{k-1}n + D_k$ (if 1 is not a characteristic root) $n^r[D_0n^k + \dots + D_{k-1}n + D_k]$ (if 1 is an r -fold characteristic root)
$\alpha^n n^k$	$[D_0n^k + \dots + D_{k-1}n + D_k]\alpha^n$ (if α is not a characteristic root) $n^r[D_0n^k + \dots + D_{k-1}n + D_k]\alpha^n$ (if α is an r -fold characteristic root)
$\cos(n\omega)$ or $\sin(n\omega)$	$D_1\cos(n\omega) + D_2\sin(n\omega)$ (if $e^{\pm j\omega}$ is not a characteristic root)
$\alpha^n \cos(n\omega)$ or $\alpha^n \sin(n\omega)$	$\alpha^n [D_1\cos(n\omega) + D_2\sin(n\omega)]$ (if $\alpha e^{\pm j\omega}$ is not a characteristic root)

➤ By observing the function form of the **free term** on the right-hand side of the equation, try to **select the form of the particular solution**.

➤ Substitute the particular solution into the equation and use the method of undetermined coefficients to determine D_i .

6.4 Solution of Linear Constant-Coefficient Difference Equations

3. Determine the Undetermined Coefficients of the Homogeneous Solution from Boundary Conditions

General Form of the Complete Solution (for Distinct Roots):

$$y(n) = C_1\alpha_1^n + C_2\alpha_2^n + \cdots + C_N\alpha_N^n + D(n) = \sum_{k=1}^N C_k\alpha_k^n + D(n)$$

C_i must be determined using N given boundary conditions. For example:

$$\left\{ \begin{array}{l} y(0) = C_1 + C_2 + \cdots + C_N + D(0) \\ y(1) = C_1\alpha_1 + C_2\alpha_2 + \cdots + C_N\alpha_N + D(1) \\ \dots\dots\dots \\ y(N-1) = C_1\alpha_1^{N-1} + C_2\alpha_2^{N-1} + \cdots + C_N\alpha_N^{N-1} + D(N-1) \end{array} \right.$$

For causal systems (where the excitation is applied at $n=0$), $y(-1), y(-2), \dots, y(-N)$ are often given as boundary conditions.

6.4 Solution of Linear Constant-Coefficient Difference Equations

Example 6-4: Find the complete solution of the difference equation

$$y(n) + 2y(n-1) = x(n) - x(n-1)$$

The excitation signal is $x(n] = n^2$, and it is known that $y(-1) = -1$.

Solution: $\alpha = -2 \quad \therefore y_h(n) = C_1(-2)^n$ Characteristic root

Free term on the right-hand side $n^2 - (n-1)^2 = 2n-1$

$y_p(n) = D_0n + D_1$ Form of Particular Solution

$$D_0n + D_1 + 2D_0(n-1) + 2D_1 = 2n-1$$

$$\begin{cases} 3D_0 = 2 \\ 3D_1 - 2D_0 = -1 \end{cases} \Rightarrow D_0 = \frac{2}{3}, \quad D_1 = \frac{1}{9}$$

Substitute the particular solution into the difference equation

$$y_p(n) = \frac{2}{3}n + \frac{1}{9}$$

Particular Solution-Forced Response

6.4 Solution of Linear Constant-Coefficient Difference Equations

Complete Solution = Homogeneous Solution + Particular Solution

$$y(n) = C_1(-2)^n + \frac{2}{3}n + \frac{1}{9}$$

Substitute the boundary condition to find the undetermined coefficient, C_1

$$y(-1) = C_1(-2)^{-1} + \frac{2}{3}(-1) + \frac{1}{9} = -1$$

$$C_1 = \frac{8}{9}$$

Obtain the closed-form of the complete solution:

$$y(n) = \frac{8}{9}(-2)^n + \frac{2}{3}n + \frac{1}{9}$$

Natural/Free Response

Forced Response

1. Repeat the examples in the lecture notes.
2. Determine whether the following sequence is a periodic signal. If yes, what is its period?

$$(1) \cos\left(\frac{3}{7}n - \frac{\pi}{8}\right)$$

$$(2) e^{j\frac{\pi}{8}n}$$

3. Given the difference equation below:

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n),$$

where $x(n) = u(n)$, $y(-1) = 5$, $y(-2) = 1$.

- (1) Solve for the total response $y(n)$.
- (2) Determine the free response $y_h(n)$ and the forced response $y_p(n)$.